

STOCK WATCHTOWER

Sources Toolkit

A companion guide for news, market data, API keys, source scoring, trending chatter and overnight market context.

Signup and key map

Create accounts only when the app reaches the milestone that needs them.

No key needed

RSS feeds

SEC public data

Stooq downloads

yfinance

HN Algolia search

Free key useful

OpenFIGI key

Finnhub key

FMP key

Twelve Data key

Companies House key

Reddit app key

Optional / later

Trading 212 API

Marketaux key

NewsAPI dev key

OpenAI API key

Stocktwits access

Purpose

This document is the toolkit. The earlier guide is the assembly manual. Use this one to decide which accounts to create, which APIs to test first, and how to keep confirmed news separate from gossip.

Prepared for a local, personal, Python + Streamlit + CSV edition of Stock Watchtower. Date checked: 7 July 2026. API limits and free tiers change often, so verify provider pages before committing to a source.

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1. Toolkit mindset

Build a watchtower, not a rumour mill.

The source system should help you collect information without flattening all information into the same trust level. A regulatory filing is evidence. A company investor relations release is evidence. A Reuters or BBC item is generally useful context. A Reddit thread, Stocktwits message, X post, Discord rumour or YouTube comment is attention, not proof.

The app should therefore keep three ideas separate: evidence, commentary and chatter. Evidence can update your thesis. Commentary can help interpret evidence. Chatter can point you towards something to investigate, but it should not be allowed to masquerade as confirmed news.

Core rule

Every item in the aggregator needs a source type, reliability tier, timestamp, URL, and note field. If the app cannot explain where an item came from, it should not appear in a decision-support signal.

Layer	What it is	How to use it
Confirmed	Official filings, RNS, company releases, regulator sources	Highest trust. Store and cite.
Media	Business media, finance APIs, RSS feeds	Good context. Verify major claims.
Chatter	Forums, Reddit, HN, Stocktwits, social posts	Trend/attention signal only. Label as unverified.
Your notes	Manual research notes and thesis comments	High value because it captures your reasoning over time.

Source reliability ladder

Use gossip for attention. Use official filings and reputable reporting for evidence.

Tier 1 - Official / regulatory

SEC EDGAR, RNS, FCA NSM, company investor relations

Tier 2 - Market data / identifiers

OpenFIGI, quote APIs, Stooq, Companies House

Tier 3 - Reputable media

BBC, Guardian, Yahoo Finance, Reuters/FT where accessible

Tier 4 - Specialist commentary

Sector blogs, newsletters, analyst notes, trade publications

Tier 5 - Community chatter

Reddit, Hacker News, Stocktwits, forums

Tier 6 - Raw social rumour

X/Twitter, Discord, Telegram, YouTube comments

2. Minimum source stack

Start narrow. Add sources only when the app needs them.

The fastest useful setup is not a giant source net. Start with a small set of dependable sources, then add specialist and chatter sources as separate tabs. This keeps the app responsive and stops the project from turning into a scraper zoo.

Phase	Source set	Why this phase exists
V1 - local dashboard	holdings.csv, manual notes, yfinance/mock quotes	Get the app useful before external dependency chaos.
V2 - identifiers	OpenFIGI and manual exchange/currency checks	Avoid wrong listings, ADR/local listing confusion and ticker collisions.
V3 - confirmed news	SEC, RNS/FCA NSM, company IR links	Ground the research feed in official evidence.
V4 - media	BBC/Guardian RSS, Yahoo Finance, FMP/Finnhub news	Add broad market context and company headlines.
V5 - chatter	Reddit, HN, Stocktwits/manual sources	Add trending gossip while keeping it clearly labelled.
V6 - overnight context	Asia indices, futures, FX, commodities	Pre-open watch signals for later-opening markets.

Recommended first signups

OpenFIGI, Finnhub or FMP, Companies House, Reddit developer app if you want chatter, and optional Twelve Data. Alpha Vantage is useful but its free limit is restrictive for a portfolio tracker.

3. Software and account checklist

What to install, what to sign up for, and what to leave until later.

You can build and test most of the source system without signing up for everything. Some sources are public, some need a free API key, and some are better treated as later/optional because their free tier is limited, their access is aimed at partners, or they become useful only after the dashboard is working.

Tool / account	Needed when	Key?	Notes
VS Code + Python extension	Immediately	No	Main editor and Python workflow. Microsoft notes that VS Code needs the Python extension and a separate Python interpreter. [S27]
Python + virtual environment	Immediately	No	Local runtime for Streamlit and scripts.
GitHub	Immediately	No	Code backup. Keep personal CSVs out of public repos.
Streamlit	Immediately	No	Local dashboard framework; Streamlit presents itself as an open-source Python framework for interactive apps. [S25]
python-dotenv	When API keys arrive	No account	Reads .env key-value pairs into environment variables. [S26]
OpenFIGI	Identifier phase	Optional/free	Free API; free key improves throughput. [S8]
Finnhub	Quote/news phase	Yes	Free personal plan available. [S9]
FMP	Quote/news phase	Yes	Basic tier shows 250 calls/day. [S10]
Twelve Data	Alternative quote phase	Yes	Basic/free plan describes API credits. [S12]

Alpha Vantage	Fallback/indicators	Yes	Free limit currently 25 requests/day. [S11]
Companies House	UK company data	Yes	API key auth required for requests. [S7]
Reddit developer app	Trending chatter	Yes	Eligible free usage has 100 QPM per OAuth client ID. [S21]
OpenAI API	Later AI/RAG only	Yes	ChatGPT Plus does not mean API access; API is separately billed. [S29]

Signup and key map

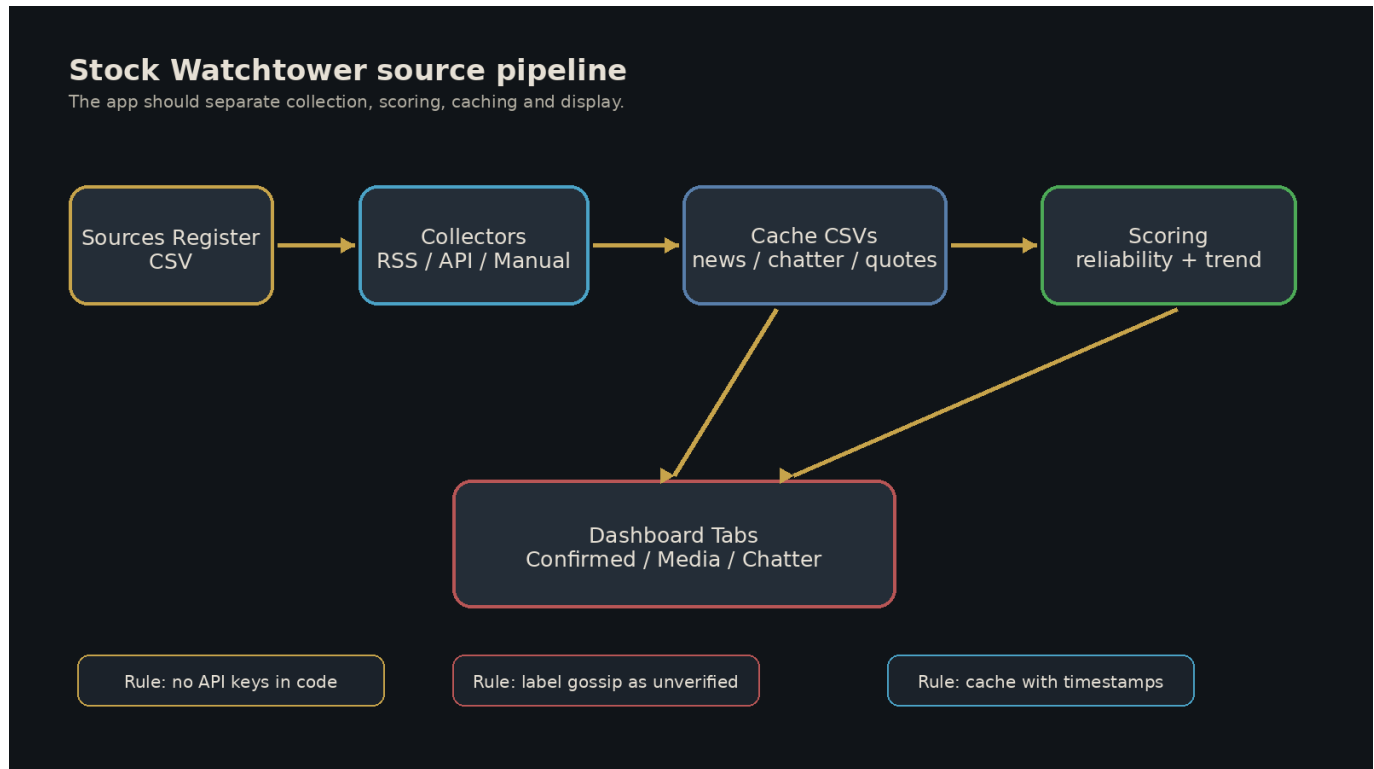
Create accounts only when the app reaches the milestone that needs them.

No key needed	Free key useful	Optional / later
RSS feeds	OpenFIGI key	Trading 212 API
SEC public data	Finnhub key	Marketaux key
Stooq downloads	FMP key	NewsAPI dev key
yfinance	Twelve Data key	OpenAI API key
HN Algolia search	Companies House key	Stocktwits access
	Reddit app key	

4. The source pipeline

A simple data flow that keeps sources clean.

The best structure is a small pipeline: a Sources Register tells the app what exists; collectors fetch or import items; cache CSVs save the results; scoring functions label quality and trend strength; dashboard tabs display the result without mixing evidence and rumour.



The pipeline is designed for local CSV persistence. You do not need hosting or a cloud database. Each cache should be safe to delete and regenerate, while data you care about - holdings, transactions, strategy profile and research notes - should be backed up before writes.

```
stock-watchtower/  
data/  
  sources_register.csv  
  holdings.csv  
  transactions.csv  
  research_notes.csv  
  strategy_profile.csv  
cache/  
  quotes_cache.csv  
  news_cache.csv  
  chatter_cache.csv  
  overnight_markets.csv  
src/  
  sources.py  
  news.py  
  chatter.py  
  source_scoring.py
```

```
quotes.py
overnight_signals.py
```

5. Sources Register CSV schema

The central ledger of where information is allowed to come from.

The Sources Register is one of the most important files in the app. It lets you add, disable, score and document sources without hardcoding them across the codebase.

```
source_id,name,category,access_method,url,reliability_tier,enabled,requires_key,api_env_var,notes
sec_edgar,SEC EDGAR,official,rss,https://www.sec.gov/search-filings,1,true,false,,US official filings
fca_nsm,FCA NSM,official>manual,https://www.fca.org.uk/markets/primary-markets/regulatory-disclosures/national-storage-
mechanism,1,true,false,,UK regulated information
openfigi,OpenFIGI,data_provider,api,https://www.openfigi.com/api/overview,2,true,true,OPENFIGI_API_KEY,Instrument
identifiers
finnhub,Finnhub,data_provider,api,https://finnhub.io/docs/api,2,false,true,FINNHUB_API_KEY,Quotes news fundamentals
guardian_business,Guardian Business,reputable_media,rss,https://www.theguardian.com/uk/business/
rss,3,true,false,,Personal non-commercial RSS use
reddit,Reddit,community_chatter,api,https://support.reddithelp.com/hc/en-us/articles/16160319875092-Reddit-Data-API-
Wiki,5,false,true,REDDIT_CLIENT_ID,Chatter only
```

Column	Meaning	Rule
source_id	Stable machine name	Never change casually after cache data exists.
category	official, media, data_provider, community_chatter, etc.	Controls which dashboard tab receives the item.
access_method	rss, api, manual, scrape, csv	Prefer rss/api/manual. Avoid scraping when possible.
reliability_tier	1 highest to 6 lowest	Do not let tier 5-6 trigger high-confidence signals.
enabled	true/false	Lets you pause noisy or broken sources.
requires_key	true/false	Prevents accidental API calls before keys exist.
api_env_var	Name of environment variable	Never store the actual key here.

6. Official and regulatory sources

The evidence layer.

These sources should sit at the top of the trust model. They are not always easy to parse, but they are the sources most likely to contain market-moving facts rather than commentary. For UK holdings, prioritise RNS, FCA NSM and company investor relations. For US holdings and ADRs, prioritise SEC EDGAR and company investor relations.

Source	Use	Access / signup	Project note
SEC EDGAR	US filings, 10-K, 10-Q, 8-K, XBRL, insider forms	No key; follow fair access, identify User-Agent, keep under SEC request limits. [S1][S3]	Use for US stocks/ADRs. Start with RSS/API metadata before full filings.
SEC structured RSS	Recent structured disclosure submissions	RSS; updated during SEC operating windows. [S2]	Good for latest filing watcher.
FCA NSM	UK regulated information storage	Manual/web first. [S4]	Use as official UK disclosure archive.
LSE/RNS	UK company regulatory announcements	Website/manual; paid/pro sources may exist. [S5]	Primary UK announcement source for LSE-traded companies.
Companies House	UK company profiles, officers, PSCs, filings metadata	Requires API key sent via basic auth for API requests. [S6][S7]	Useful for company background, not market price.
Company investor relations	Results, presentations, trading updates, webcasts	Manual/RSS/email alerts depending on company	Add company-specific IR URLs per holding.
Investigate / other RNS browsers	Public announcement browsing	Manual/web	Convenient aggregator; treat as secondary to official RNS/NSM.

Implementation advice

For the first pass, store official source URLs manually against each holding. Do not try to parse every filing type immediately. Add one source at a time and cache the result.

```
# Example official_links.csv
holding_id,ticker,source_type,name,url,notes
h001,RR.L,investor_relations,Rolls-Royce IR,https://www.rolls-royce.com/investors.aspx,Company releases and results
h001,RR.L,rns,LSE RNS search,https://www.londonstockexchange.com/news?tab=news-explorer,Search ticker manually
h002,NOK,sec,SEC EDGAR,https://www.sec.gov/search-filings,US ADR filings and issuer materials
```

7. Market data and identifiers

Prices are not all the same thing.

Ticker alone is not enough. The same company can have a local listing, ADR, Xetra listing, London listing, ETF share class, different currency line, or similar ticker collision. The app should store listing details and use identifiers where possible.

Source	Best use	Key/signup	Cautions
yfinance	Fast local prototyping and research	No key	Unofficial; intended for research/education and not affiliated with Yahoo. [S14]
OpenFIGI	FIGI/identifier mapping and instrument search	Optional free key	Excellent for avoiding wrong-listing problems. [S8]

Finnhub	Quotes, fundamentals, news, alternative data	Free API key	Personal plan; check current limits and licensing. [S9]
FMP	Quotes, financials, stock news	Free API key	Basic docs show 250 calls/day; plan/timeframe varies. [S10]
Alpha Vantage	Historical data, indicators, forex, crypto, fundamentals	Free key	25 requests/day on free tier is tight for portfolio polling. [S11]
Twelve Data	Global quotes/forex/crypto, API/CSV output	Free key	Credit-based plan; verify current market coverage. [S12]
Stooq	Free historical CSV downloads	No key	Good fallback for historical data, not a polished live quote API. [S13]
Nasdaq Data Link	Economic/financial datasets	Free account for some datasets	Many datasets are premium; free/open data exists. [S15]
Trading 212 API	Your own account positions/orders if you choose	Account API key	Optional later; keep trading/execution separate from research app. [S24]

Recommended provider order

Use yfinance or mock prices while building. Add OpenFIGI for identifiers. Then choose one proper quote/news provider - Finnhub or FMP - and build a provider interface so it can be swapped later.

```
# Suggested quote object
symbol: str
exchange: str | None
currency: str
price: float
previous_close: float | None
change_percent: float | None
source: str
fetched_at: str
stale: bool = False
```

8. Reputable media and RSS news

Good context, but still not the same as filings.

RSS is a simple, open way to pull headlines from sites that publish feeds. The RSS 2.0 specification describes RSS as an XML web content syndication format. [S16] For a local personal app, RSS is usually the best first news layer because it avoids API key complexity and reduces scraping.

Source	Use	Access	Notes
BBC Business	UK/global business context	RSS feed examples under BBC feed terms. [S17]	Good broad context, not ticker-perfect.
Guardian Business	UK business/economy context	RSS for personal non-commercial purposes. [S18]	Good UK macro/economy/company context.
Yahoo Finance	Ticker pages and headlines	Website/yfinance adjacent	Useful manual context and headlines.
GDELT	Global news/event monitoring	Open data/API style access. [S19]	Powerful but noisy; use later for trend detection.
FMP/Finnhub news	Ticker-oriented news API	API key	Good when you want company-specific headlines linked to tickers.
Marketaux	Finance news/sentiment API	API key	Useful later for entity-linked sentiment; verify current free tier.
NewsAPI	General news API	Developer key	Developer plan is for development/testing, not production/internal production use. [S20]
Reuters/FT	High-quality professional news	Mostly manual/subscription/paywall	Use as reading sources; do not assume free API/RSS access.

```
# Suggested news_cache.csv
```

```
item_id,ticker,source_id,source_type,title,url,published_at,summary,sentiment_hint,reliability_tier,fetched_at,notes
n001,RR.L,guardian_business,reputable_media,"Defence shares move after...",https://...,2026-07-07T08:00:00,"Short
summary",neutral,3,2026-07-07T09:00:00,"Read full article before acting"
```

9. Trending gossip and chatter

Early smoke, not confirmed fire.

A Trending Gossip panel is useful, but only if it is framed correctly. It should answer: what are people suddenly talking about, where are they saying it, and is there any hard evidence? It should not answer: what should I buy?

Source	Best use	Access / signup	Caution
Reddit	Retail sentiment, due diligence threads, rumour detection	OAuth app; eligible free usage limit is 100 QPM per client ID. [S21]	Noisy, hype-prone, brigading possible.
Hacker News	Tech/startup/company discussion and AI/software chatter	Public Algolia API. [S22]	Tech bias; not finance-specific.
Stocktwits	Ticker-level sentiment/chatter	API/docs exist; partner/enterprise orientation may apply. [S23]	Very noisy; momentum/hype heavy.
X/Twitter	Fast rumours and analyst chatter	Manual initially	API access changes often; high bot/noise risk.
YouTube/podcasts	Narrative tracking	Manual/RSS if available	Often slow, promotional or

Specialist forums/Substack	Sector expertise	Manual/RSS/email	personality-driven. Quality varies hugely; classify carefully.
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Display rule

Trending Gossip must have a visible unverified label unless it links to official evidence or reputable reporting. It can trigger a watch signal, not a conviction signal.

```
# Suggested chatter_cache.csv
item_id,ticker,source_id,source_type,title,url,published_at,mention_count,engagement_score,sentiment_hint,attention_score,evidence_score,confidence_score,rumour_flag,notes
c001,RR.L,reddit,community_chatter,"SMR contract speculation thread",https://...,2026-07-07T08:30:00,5,42,bullish,72,10,18,true,"No official RNS found yet"
```

10. Source scoring and safety rules

Score attention separately from truth.

Do not make one magic score. A rumour can be very visible and still very weak. A quiet RNS can be boring but highly reliable. The app should separate trend strength from confidence.

Score	Meaning	High value means
attention_score	How much people are talking about it	The item/topic is visible or growing.
novelty_score	How unusual the current chatter is	Something changed versus normal baseline.
source_quality_score	How trustworthy this source type usually is	Official/media sources score higher than forums.
evidence_score	Whether item links to filings/data/reputable reporting	The claim has support outside social chatter.
confidence_score	Source quality plus evidence	The item deserves more trust.

```
def score_item(source_tier: int, has_official_link: bool, has_reputable_link: bool, engagement: int, recent_mentions: int, baseline_mentions: int) -> dict:
    # Lower tier number means better source. Convert 1..6 into a quality score.
    source_quality_score = max(0, 100 - ((source_tier - 1) * 18))

    evidence_score = 0
    if has_official_link:
        evidence_score += 60
    if has_reputable_link:
        evidence_score += 30

    attention_score = min(100, engagement * 2 + recent_mentions * 5)
    novelty_score = min(100, max(0, recent_mentions - baseline_mentions) * 10)

    confidence_score = min(100, int((source_quality_score * 0.55) + (evidence_score * 0.45)))
    trend_score = min(100, int((attention_score * 0.6) + (novelty_score * 0.4)))
```

```
return {  
  "attention_score": attention_score,  
  "novelty_score": novelty_score,  
  "trend_score": trend_score,  
  "confidence_score": confidence_score,  
}
```

Signal wording

Use phrases such as "watch", "verify", "possible read-through", "unconfirmed chatter" and "evidence needed".
Avoid direct buy/sell/hold commands.

11. API key and secret management

Keys go in .env, not in code and not in GitHub.

For a local app, .env is the simplest secret store. python-dotenv reads key-value pairs from a .env file and can set them as environment variables. [S26] Do not commit .env or personal data CSVs to a public repository. Keep example files instead.

```
# .env.example
OPENFIGI_API_KEY=
FINNHUB_API_KEY=
FMP_API_KEY=
ALPHA_VANTAGE_API_KEY=
TWELVE_DATA_API_KEY=
COMPANIES_HOUSE_API_KEY=
REDDIT_CLIENT_ID=
REDDIT_CLIENT_SECRET=
REDDIT_USER_AGENT=StockWatchtowerLocal/0.1 by yourname
MARKETAUX_API_KEY=
NEWSAPI_KEY=
OPENAI_API_KEY=
TRADING212_API_KEY=
```

```
# .gitignore
.env
.venv/
__pycache__/
*.pyc

# Personal finance data
data/*.csv
cache/*.csv
backups/
exports/

# Keep templates/examples
!data/example_holdings.csv
!data/example_sources_register.csv
!data/example_strategy_profile.csv
```

```
# src/config.py
import os
from dotenv import load_dotenv

load_dotenv()
```

```
def get_key(name: str) -> str | None:
    value = os.getenv(name)
    if value and value.strip():
        return value.strip()
    return None

FINNHUB_API_KEY = get_key("FINNHUB_API_KEY")
FMP_API_KEY = get_key("FMP_API_KEY")
OPENFIGI_API_KEY = get_key("OPENFIGI_API_KEY")
```

Local security habit

Keep .env local. Keep your GitHub repo private while experimenting, or use .gitignore aggressively. Never paste API keys into ChatGPT, GitHub issues, screenshots or code examples.

12. Overnight markets and Asian session read-through

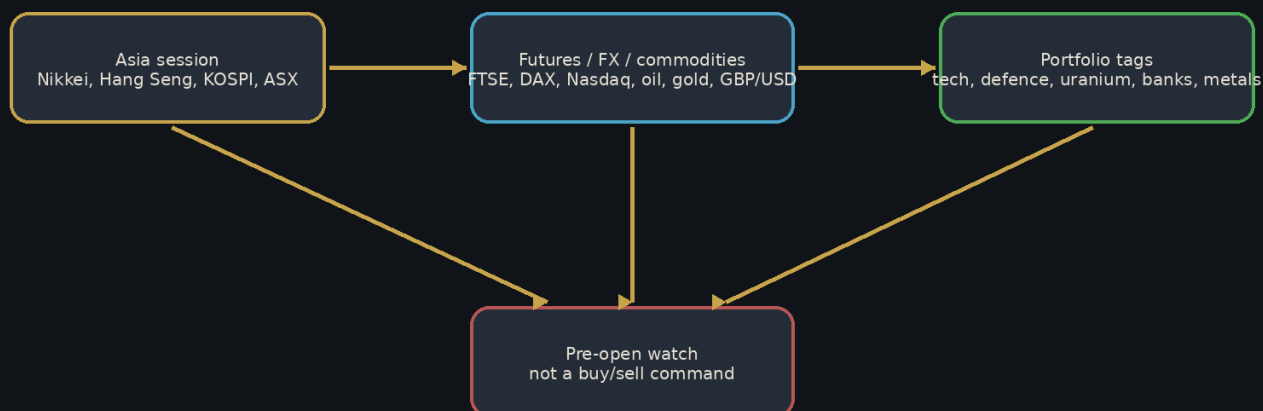
Useful context, not a crystal ball.

Asian market movement can be useful because the global trading day moves from Asia/Australia into Europe and then into the US. Fidelity describes Asian markets as opening first, followed by Europe and then the US, with Europe still open for the first part of the US session. [S28]

The mistake is to reduce this to one index. A weak Nikkei alone does not mean the FTSE or Nasdaq will open weak. A better signal combines Asian indices, US and European futures, FX, commodities, bond yields and sector read-through.

Overnight signal flow

Asian moves can be context for later opens, but futures and sector read-through matter more than one index alone.



Input	Why it matters	Example signal
Nikkei 225	Japan/exporters/global risk mood	Japan risk-off tone.
Hang Seng / CSI 300	China/HK risk, property, banks, miners, luxury	China weakness may affect miners and emerging markets.
KOSPI / Taiwan tech	Semiconductor read-through	Weak chip session may affect AI/semiconductor holdings.
FTSE/DAX/Euro Stoxx futures	Direct pre-open Europe context	European futures imply likely open direction.
S&P/Nasdaq futures	US/global growth risk	Tech/growth exposure may be pressured.
FX: GBP/USD, EUR/USD, USD/JPY	Currency translation and macro stress	GBP jump may affect overseas revenue translation.
Oil/gold/copper	Energy, inflation, industrial demand, risk appetite	Oil spike may pressure airlines and support energy.

Signal wording example

"Asia risk-off plus Nasdaq futures weakness. Relevant tags: technology, AI, semiconductors. Treat as pre-open watch context; check company-specific news before action."

13. Implementation roadmap

Add sources in controlled waves.

1. Create data/sources_register.csv and add only manual/RSS/no-key sources.
2. Add source loading and validation functions in src/sources.py.
3. Add BBC/Guardian RSS parsing with feedparser and cache to cache/news_cache.csv.
4. Add official links per holding: IR, RNS/NSM, SEC/EDGAR where relevant.
5. Add one market data provider interface with yfinance/mock fallback.
6. Add OpenFIGI lookup for instrument accuracy.
7. Add one real quote/news API key provider, preferably Finnhub or FMP.
8. Add Trending Gossip tab with HN first, then Reddit OAuth.
9. Add source scoring and visible labels.
10. Add overnight market context after quote/news caching is stable.
11. Only then consider AI summaries or RAG; manual research packets are enough at first.

Milestone prompt for ChatGPT

Add the Sources Register milestone to Stock Watchtower Local CSV Edition.

Requirements:

- Create data/sources_register.csv with example rows.
- Create src/sources.py.
- Load and validate sources.
- Required columns: source_id, name, category, access_method, url, reliability_tier, enabled, requires_key, api_env_var, notes.
- Show an app Settings/Sources section listing enabled/disabled sources.
- Do not fetch news yet.
- Do not add API keys to code.
- Keep the implementation local and CSV-based.

14. Copy-paste templates

Blank canvases for the source toolkit.

These templates match the scaffold-first workflow: create the blank files first, then fill them as milestones arrive.

```
# data/example_sources_register.csv
source_id,name,category,access_method,url,reliability_tier,enabled,requires_key,api_env_var,notes
sec_edgar,SEC EDGAR,official,rss,https://www.sec.gov/search-filings,1,true,false,,US official filings and APIs
fca_nsm,FCA NSM,official>manual,https://www.fca.org.uk/markets/primary-markets/regulatory-disclosures/national-storage-
mechanism,1,true,false,,UK regulated information storage
lse_rns,LSE RNS,official>manual,https://www.londonstockexchange.com/news,1,true,false,,UK regulatory news
announcements
companies_house,Companies House,data_provider,api,https://developer.company-
information.service.gov.uk/,2,false,true,COMPANIES_HOUSE_API_KEY,UK company data
openfigi,OpenFIGI,data_provider,api,https://www.openfigi.com/api/overview,2,false,true,OPENFIGI_API_KEY,Identifier
mapping
bbc_business,BBC Business,reputable_media,rss,https://feeds.bbc.co.uk/news/business/rss.xml,3,true,false,,Business RSS
reddit,Reddit,community_chatter,api,https://www.reddit.com/dev/api/,5,false,true,REDDIT_CLIENT_ID,Chatter only
hacker_news,Hacker News,community_chatter,api,https://hn.algolia.com/api,5,true,false,,Tech/startup chatter
```

```
# src/sources.py - starter skeleton
from dataclasses import dataclass
from pathlib import Path
import pandas as pd

SOURCES_PATH = Path("data/sources_register.csv")
REQUIRED_COLUMNS = {
    "source_id", "name", "category", "access_method", "url",
    "reliability_tier", "enabled", "requires_key", "api_env_var", "notes"
}

@dataclass
class SourceRecord:
    source_id: str
    name: str
    category: str
    access_method: str
    url: str
    reliability_tier: int
    enabled: bool
    requires_key: bool
    api_env_var: str | None
    notes: str | None

def load_sources(path: Path = SOURCES_PATH) -> pd.DataFrame:
    df = pd.read_csv(path)
    missing = REQUIRED_COLUMNS - set(df.columns)
```

```

if missing:
    raise ValueError(f'sources_register.csv is missing columns: {sorted(missing)}')
df["reliability_tier"] = pd.to_numeric(df["reliability_tier"], errors="raise").astype(int)
return df

```

```

def enabled_sources(df: pd.DataFrame) -> pd.DataFrame:
    return df[df["enabled"].astype(str).str.lower().isin(["true", "1", "yes"])]

```

```

# src/news.py - RSS starter skeleton
from pathlib import Path
from datetime import datetime, timezone
import pandas as pd
import feedparser

CACHE_PATH = Path("cache/news_cache.csv")

def fetch_rss(source_id: str, url: str, reliability_tier: int) -> pd.DataFrame:
    feed = feedparser.parse(url)
    rows = []
    for entry in feed.entries:
        rows.append({
            "item_id": f'{source_id}:{entry.get("id") or entry.get("link")}',
            "ticker": "",
            "source_id": source_id,
            "source_type": "rss",
            "title": entry.get("title", ""),
            "url": entry.get("link", ""),
            "published_at": entry.get("published", ""),
            "summary": entry.get("summary", ""),
            "sentiment_hint": "unknown",
            "reliability_tier": reliability_tier,
            "fetched_at": datetime.now(timezone.utc).isoformat(),
            "notes": "",
        })
    return pd.DataFrame(rows)

```

15. References

Official pages and documentation checked while preparing this toolkit.

- S1. SEC Search Filings / EDGAR APIs and RSS feeds:** <https://www.sec.gov/search-filings>
- S2. SEC Structured Disclosure RSS feeds:** <https://www.sec.gov/data-research/structured-data/structured-disclosure-rss-feeds>
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